

On the category of metric compact Hausdorff spaces

See recording at <https://youtu.be/7thZXynhztU>

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Based on:



Marco Abbadini, Dirk Hofmann.

Barr-coexactness for metric compact Hausdorff spaces.

Theory and Applications of Categories (2025). [Link](#).

A category

- ▶ Separated metric compact Hausdorff spaces (compact Hausdorff space + **metric**).

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Metric is similar to Order (quantale-enrichment).

(Quantales are special symmetric strict monoidal categories.)

Historical Background

Compact Hausdorff spaces = Eilenberg-Moore algebras for the ultrafilter monad on **Set**.

CompHaus is a variety of infinitary algebras; it has an algebraic flavor. For example, it is Barr-exact.

CompHaus has also a topological flavor.

Usually, the opposite of a category with a topological flavor has an algebraic flavor:

- ▶ While **Top** is not regular, **Top**^{op} is regular (in fact, a quasivariety of infinitary algebras [Barr, Pedicchio, 1995]).
- ▶ The opposite of the category of Stone spaces is a variety of algebras (Boolean algebras) [Stone, 1936]:

$$\mathrm{hom}_{\mathbf{Stone}}(-, \{0, 1\}): \mathbf{Stone}^{\mathrm{op}} \rightarrow \mathbf{Set}$$

is monadic.

(We recall that every monadic functor to **Set** is representable, by the free algebra on one generator.)

Theorem ([Duskin, 1969], all details in [Barr, Wells, 1985])

The functor

$$\mathrm{hom}_{\mathbf{CompHaus}}(-, [0, 1]): \mathbf{CompHaus}^{\mathrm{op}} \rightarrow \mathbf{Set}$$

is monadic.

$\mathbf{CompHaus}^{\mathrm{op}}$ is a variety of infinitary algebras: it has an algebraic flavor.

$\mathbf{CompHaus}$ is complete, cocomplete, and Barr-coexact, $[0, 1]$ is regular injective and a regular cogenerator (= Urysohn's lemma).

Stone duality (between Stone spaces and Boolean algebras) has an important generalization to ordered-topological spaces: Priestley duality.

Priestley space := Stone space + compatible partial order.

Priestley duality [Priestley, 1970]: Priestley spaces are dual to bounded distributive lattices (which form a variety).

$$\mathrm{hom}_{\mathbf{Priestley}}(-, \{0, 1\}): \mathbf{Priestley}^{\mathrm{op}} \rightarrow \mathbf{Set}$$

is monadic.

Compact Hausdorff CompHaus^{op} $\xrightarrow{\text{hom}(-,[0,1])}$ Set	Stone Stone^{op} $\xrightarrow{\text{hom}(-,\{0,1\})}$ Set
Compact Hausdorff + order ?	Stone + order Priestley^{op} $\xrightarrow{\text{hom}(-,\{0,1\})}$ Set

Question [Hofmann, Neves, Nora, 2018]: is there an analogue with “*compact Hausdorff* spaces + order” instead of “*Stone* spaces + order”?

Nachbin space (a.k.a. compact ordered space) [Nachbin, 1948]: compact Hausdorff space + compatible partial order.

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Theorem (A., 2019)

$$\mathrm{hom}_{\mathbf{Nachbin}}(-, [0, 1]): \mathbf{Nachbin}^{\mathrm{op}} \rightarrow \mathbf{Set}$$

is monadic.

(See [A., Reggio, 2019] for a nicer proof.)

It is also a way to collect categorical properties of **Nachbin**:

1. **Nachbin** is complete and cocomplete [Tholen, 2009].
2. $[0, 1]$ is a regular injective regular cogenerator [Nachbin, 1960].
3. **Nachbin** is Barr-coexact [A., Reggio, 2020]. (Coregularity and something more already in [Hofmann, Neves, Nora, 2018]).

Separated metric compact Hausdorff spaces

From Lawvere, we know that order is similar to metric.
Is there an analogue in the metric setting?

Before recalling **separated metric compact Hausdorff spaces**, let us see some drawbacks of the category of classical **compact metric spaces** and non-expansive maps:

- ▶ **not cocomplete.**

Remedy: allow distance ∞ .

- ▶ **not complete.**

Remedy: topology **compatible** with the metric, rather than **induced** by it.

Definition

A *metric* on a set X is a map $d: X \times X \rightarrow [0, \infty]$ satisfying:

- ▶ (reflexivity) $d(x, x) = 0$;
- ▶ (triangle inequality) $d(x, z) \leq d(x, y) + d(y, z)$.

A metric is *separated* if $d(x, y) = 0 = d(y, x)$ implies $x = y$.

All results are true also when restricted to the symmetric case ($d(x, y) = d(y, x)$.)

If d is only allowed to take values 0 and ∞ ,

- ▶ metric = preorder (where $d(x, y) = 0$ means $x \leq y$),
- ▶ separated metric = partial order.

Definition ([Hofmann, Reis, 2018])

(Separated) metric compact Hausdorff space $\coloneqq$ compact Hausdorff space X equipped with a lower semicontinuous (separated) metric $X \times X \rightarrow [0, \infty]$.

Lower semicontinuous:

$$d(x_0, y_0) \leq \liminf_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} d(x, y).$$

I.e.: small topological perturbations may yield great increments in distances, but not great decrements.

Equivalently, continuous wrt the topology generated by the sets $(a, \infty]$.

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Example: any compact metric space (in the classical sense).

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Example: $[0, \infty]$, with $d(a, b) = \begin{cases} b - a & \text{if } a < b; \\ 0 & \text{otherwise.} \end{cases}$

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Example: For any set X , $[0, 1]^X$ with the product metric (= sup metric) and the product topology.

Metric compact Hausdorff spaces are the algebras for the “metric ultrafilter” monad on the category of metric spaces and nonexpansive maps (See [Hofmann, Reis, 2018], building on [Tholen, 2009]).

MetCH_{sep} := category of separated metric compact Hausdorff spaces and nonexpansive continuous maps.

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Question

Is

$$\text{hom}_{\mathbf{MetCH}_{\text{sep}}}(-, [0, \infty]): \mathbf{MetCH}_{\text{sep}}^{\text{op}} \rightarrow \mathbf{Set}$$

monadic?

I.e.:

1. Is $\mathbf{MetCH}_{\text{sep}}$ complete and cocomplete?
2. Is $\mathbf{MetCH}_{\text{sep}}$ Barr-coexact?
3. Is $[0, \infty]$ a regular injective regular cogenerator of $\mathbf{MetCH}_{\text{sep}}$?

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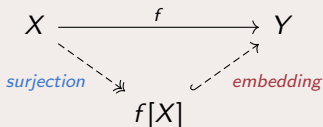
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? (*Future work.*)

Theorem ([A., Hofmann, 2025])

1. $\mathbf{MetCH}_{\text{sep}}$ has (*epi*, *regular mono*) factorization:



equip $f[X]$ with the metric and topology induced by Y

2. *epis* = *surjective morphisms*;
3. *regular monos* = *strong monos* = *embeddings*.

Theorem ([A., Hofmann, 2025])

MetCH_{sep} is coregular.

$$\begin{array}{ccc} K & \hookrightarrow & X \\ f \downarrow & \lrcorner & \downarrow \\ K' & \hookrightarrow & X' \end{array}$$

Given an embedding $K \hookrightarrow X$ and a morphism $f: K \rightarrow K'$, inside X we can replace K by a copy of K' (and appropriate adjustments outside of K' induced by f).

Theorem ([A., Hofmann, 2025])

MetCH_{sep} is *Barr-coexact*.

For a separated metric compact Hausdorff space X , a closed subset $K \subseteq X$ induces a quotient of $X + X$ by gluing the two copies of K .

Barr-coexactness: every surjective morphism $X + X \twoheadrightarrow Z$ satisfying coreflexivity, cosymmetry, cotransitivity (first-order conditions) arises in this way.

To sum up

MetCH_{sep} := category of separated metric compact Hausdorff spaces and continuous non-expansive maps. [Hofmann, Reis, 2018]

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Thank you.